MOCKITO HANDS-ON EXERCISES

EXERCISE 1: MOCKING AND STUBBING

ExternalApi.java

```
public interface ExternalApi {
   String getData();
}
```

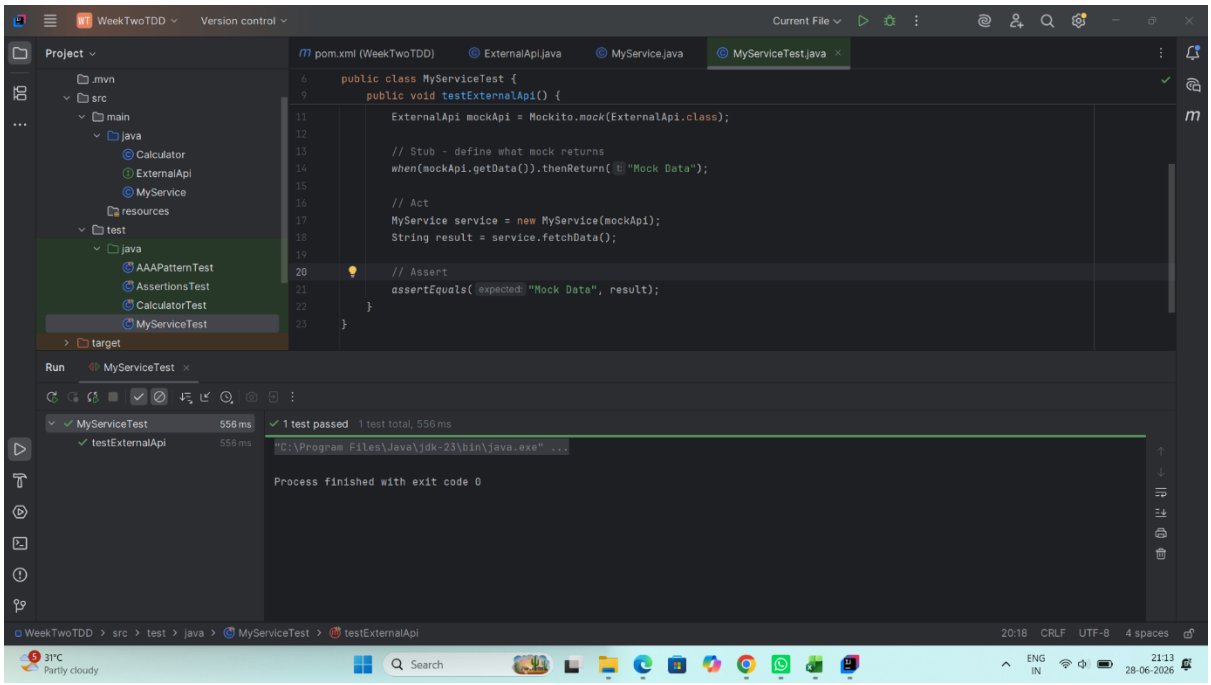
MyService.java

```
public class MyService {
   ExternalApi api;
   public MyService(ExternalApi api) {
      this.api = api;
   }
   public String fetchData() {
      return api.getData();
   }
}
```

MyServiceTest.java

```
import org.junit.Test;
import org.mockito.Mockito;
import static org.mockito.Mockito.*;
import static org.junit.Assert.*;
public class MyServiceTest {
   @Test
   public void testExternalApi() {
      // Arrange - Create mock
      ExternalApi mockApi = Mockito.mock(ExternalApi.class);
      // Stub - define what mock returns
      when(mockApi.getData()).thenReturn("Mock Data");
      // Act
      MyService service = new MyService(mockApi);
      String result = service.fetchData();
      // Assert
      assertEquals("Mock Data", result);
   }
}
```

OUTPUT


WeekTwoTDD
Version control
Current File
Project
pom.xml (WeekTwoTDD)
ExternalApi.java
MyService.java
MyServiceTest.java
.mvn
src
main
java
Calculator
ExternalApi
MyService
resources
test
java
AAAPatternTest
AssertionsTest
CalculatorTest
MyServiceTest
target
public class MyServiceTest {
public void testExternalApi() {
ExternalApi mockApi = Mockito.mock(ExternalApi.class);
// Stub - define what mock returns
when(mockApi.getData()).thenReturn("Mock Data");
// Act
MyService service = new MyService(mockApi);
String result = service.fetchData();
// Assert
assertEquals(expected: "Mock Data", result);
}
}
Run
MyServiceTest
MyServiceTest 556 ms
testExternalApi 556 ms
1 test passed 1 test total, 556 ms
"C:\Program Files\Java\jdk-23\bin\java.exe" ...
Process finished with exit code 0
WeekTwoTDD › src › test › java › MyServiceTest › testExternalApi
20:18 CRLF UTF-8 4 spaces
31°C
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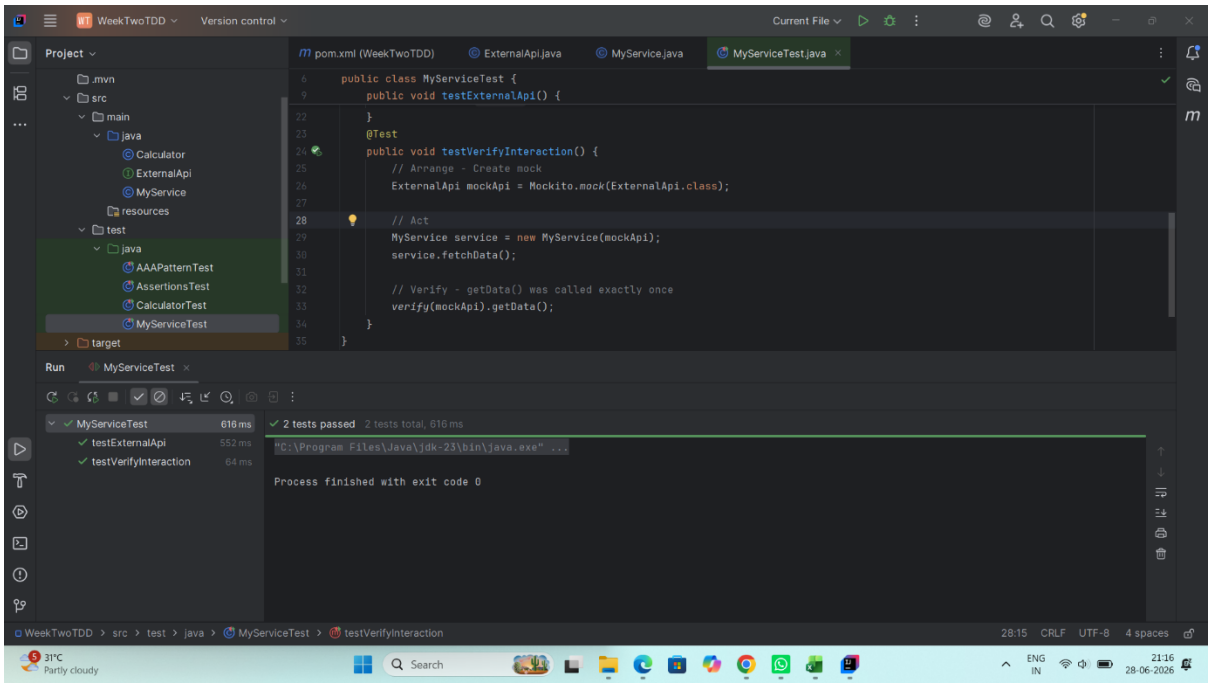
Search
ENG
IN
21:13
28-06-2026

EXERCISE 2: VERIFYING INTERACTIONS

MyServiceTest.java

```
import org.junit.Test;
import org.mockito.Mockito;
import static org.mockito.Mockito.*;
import static org.junit.Assert.*;
public class MyServiceTest {
   @Test
   public void testExternalApi() {
      ExternalApi mockApi = Mockito.mock(ExternalApi.class);
      when(mockApi.getData()).thenReturn("Mock Data");
      MyService service = new MyService(mockApi);
      String result = service.fetchData();
      assertEquals("Mock Data", result);
   }
   @Test
   public void testVerifyInteraction() {
      ExternalApi mockApi = Mockito.mock(ExternalApi.class);
      MyService service = new MyService(mockApi);
      service.fetchData();
      verify(mockApi).getData();
   }
```

OUTPUT


WeekTwoTDD
Version control
Current File
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pom.xml (WeekTwoTDD)
ExternalApi.java
MyService.java
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.mvn
src
main
java
Calculator
ExternalApi
MyService
resources
test
java
AAAPatternTest
AssertionsTest
CalculatorTest
MyServiceTest
target
public class MyServiceTest {
public void testExternalApi() {
}
@Test
public void testVerifyInteraction() {
// Arrange - Create mock
ExternalApi mockApi = Mockito.mock(ExternalApi.class);
// Act
MyService service = new MyService(mockApi);
service.fetchData();
// Verify - getData() was called exactly once
verify(mockApi).getData();
}
}
Run
MyServiceTest
MyServiceTest 616 ms
testExternalApi 552 ms
testVerifyInteraction 64 ms
2 tests passed 2 tests total, 616 ms
"C:\Program Files\Java\jdk-23\bin\java.exe" ...
Process finished with exit code 0
WeekTwoTDD › src › test › java › MyServiceTest › testVerifyInteraction
28:15 CRLF UTF-8 4 spaces
31°C
Partly cloudy
Search
ENG
IN
21:16
28-06-2026